Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – VIII (Batch 2024-2028)

Semester	Category code as per NEP	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 8	Project	PRO	24PRBT452	Major Project	0	0	24	24	12
			24PRBT453	Comprehensive Project					

Major Project

24PRBT452					Major Project					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	24	12	24	-	-	-	50	50	100

COURSE OBJECTIVES

1. To design and execute independent biotechnology research projects
2. To demonstrate effective scientific communication skills in both written and oral formats
3. To critically evaluate the potential of biotechnology to address global challenges in healthcare, agriculture, and industry.
4. To master laboratory techniques essential for biotechnology research

1. Medical Biotechnology:
   - Drug development and discovery: Identify new drug targets, develop novel therapeutics (small molecules, antibodies, gene therapies) for diseases, and optimize drug delivery systems.
   - Diagnostics: Use tools like PCR, biosensors, and next-generation sequencing to design rapid and sensitive diagnostics for infectious diseases, cancers, and genetic disorders.
   - Regenerative medicine: Focus on tissue engineering, stem cell therapies, and biomaterial development for repairing or replacing damaged organs and tissues.
   - Personalized medicine: Use genomics, proteomics, and bioinformatics to tailor diagnosis and treatment strategies to an individual's genetic makeup.
   - Biopharmaceutical Development: Scale-up production of biologics such as vaccines, monoclonal antibodies, and biosimilars using recombinant DNA technology. Explore fermentation and cell culture techniques for high-yield production.
   - Nanomedicine: Develop nanotechnology-based drug delivery systems for targeted therapies. Design nanoparticles for imaging, diagnosis, and treatment of diseases like cancer and cardiovascular disorders.
2. Bioinformatics:
   - Molecular Modelling and Drug Design: Focuses on using machine learning and deep learning algorithms to predict the three-dimensional structure of proteins, vital for understanding protein function and advancing drug design.
   - Disease Association Studies: Targets the identification of genetic variants linked to diseases by analyzing large-scale genomic data, aiming to uncover the genetic basis of complex diseases and improve diagnostics and therapies.
   - Multi-Omics Integration: Involves integrating genomics, transcriptomics, proteomics, and other 'omics' data to gain a comprehensive understanding of biological systems, aiding in the discovery of new biomarkers and therapeutic targets.

- Computational Systems Biology: Modeling and simulating biological systems using mathematical modeling, network analysis, and high-throughput data to decipher the mechanisms of biological processes and pathways.

3. Agricultural Biotechnology:
    - Crop improvement: Enhance crop yields, stress tolerance (drought, salinity), and nutritional value of crops through genetic engineering or CRISPR-based gene editing.
    - Pest and disease management: Develop bio pesticides and disease-resistant crops to reduce reliance on chemical inputs.
    - Sustainable agriculture: Use microbes for nutrient optimization, study plant-microbe interactions, and develop strategies to improve soil health.
4. Industrial Biotechnology:
    - Biofuel production: Develop microbial systems for converting biomass into bioethanol, biodiesel, and other biofuels.
    - Bioremediation: Harness the power of microbes or enzymes to clean up contaminated environments (oil spills, polluted water).
    - Bio-based materials production: Create sustainable alternatives to plastics, textiles, and other materials using bio-resources.
5. Environmental Biotechnology:
    - Wastewater treatment: Design innovative microbial systems for effective treatment of industrial and municipal wastewater.
    - Biomonitoring: Biosensors and microorganisms are used to assess environmental quality and detect pollutants.
    - Biodiversity conservation: Apply biotechnology tools for analyzing species diversity and developing conservation strategies.

6. Additional Research Areas

- Food Biotechnology: Improving food safety, preservation, nutritional value, flavor profiles, waste reduction
- Marine Biotechnology: Exploring marine organisms and their bioactive compounds for pharmaceutical and industrial applications and understanding ocean ecosystems.
- Forensic Biotechnology: Using DNA analysis and other biotechnological tools in criminal investigations.

COURSE OUTCOMES

CO1: Design and execute independent biotechnology research projects addressing challenges in healthcare, agriculture, and industry.

CO2: Demonstrate proficiency in laboratory techniques and computational tools for advanced biotechnology research.

CO3: Apply bioinformatics approaches such as molecular modeling, multi-omics integration, and systems biology to solve biological problems.

CO4: Develop innovative solutions in medical biotechnology, including drug discovery, diagnostics, and personalized medicine.

CO5: Utilize biotechnology tools to improve agricultural productivity, environmental sustainability, and industrial processes.

CO6: Communicate scientific findings effectively through written reports, oral presentations, and collaborative discussions.